Use together with the General sheet (Sheet 1).

Record measured outcomes and empirical findings reported in RCTs, pilots, field studies, implementation studies, and other real-world evaluations identified in Sheet 1.

Variables to extract

1 Study / Source

Record the study or source identifier used in Sheet 1. Distinguish separate deployments when one source reports more than one intervention.

2 AI Tool / Program

Record the AI tool or programme name consistently with Sheet 1.

3 Outcome Domain

Definition: Standardized category describing the type of outcome measured. Use one of the following categories whenever possible.

Adoption

Whether eligible users or organizations begin using the AI tool.

Examples:

- take-up rate,
- activation rate,
- share of schools adopting,
- share of teachers starting use.

Keywords / cues:
adopted, take-up, enrolled, activated, started using, participation rate

---

Usage

How much, how often, or how long the AI tool is actually used after access/adoption.

Examples:

- sessions per week,
- minutes used,
- number of interactions,
- attendance,
- continued use,

- retention.

Keywords / cues:
usage, frequency, time on platform, sessions, attendance, logins, engagement, retention

---

Learning

Changes in student knowledge, skills, test scores, grades, or academic performance.

Examples:

- mathematics test score,
- language score,
- course grade,
- standardized learning outcome.

Keywords / cues:
achievement, test score, learning gain, exam, grade, assessment score

---

Engagement

Student participation, motivation, effort, interest, or interaction with learning.

Examples:

- motivation,
- participation,
- effort,
- persistence,
- engagement index.

Keywords / cues:
engagement, motivation, participation, effort, interest

Attendance

School or class attendance outcomes.

Examples:

- attendance rate,
- absenteeism,
- days present.

Keywords / cues:
attendance, absenteeism, school days

Dropout / Retention

Whether students remain in education, complete courses, or drop out.

Examples:

- dropout rate,
- retention,
- course completion,
- persistence.

Keywords / cues:
dropout, retention, completion, persistence, withdrawal

Teacher Practice

Changes in teaching behavior or instructional practices.

Examples:

- questioning quality,
- feedback practices,
- lesson design,
- use of student ideas.

Keywords / cues:
teacher behavior, instructional practice, questioning, feedback, pedagogy

---

Teacher Productivity

Changes in teacher time, workload, output, or efficiency.

Examples:

- lesson preparation time,
- grading time,
- students served per teacher.

Keywords / cues:
time saved, workload, productivity, efficiency, preparation time

---

Access

Whether the intervention increases access to education, tutoring, feedback, or educational services.

Examples:

- number of students reached,
- access to tutoring,
- access in rural areas.

Keywords / cues:
access, reach, coverage, availability, underserved

Equity

Differences in adoption, usage, or outcomes across groups.

Examples:

- low-income vs high-income students,
- girls vs boys,
- rural vs urban,
- low-performing vs high-performing students.

Keywords / cues:
heterogeneous effects, subgroup, disadvantaged, equity, inequality

---

Inclusion / Accessibility

Outcomes specifically related to disability, special educational needs, language access, or inclusion.

Examples:

- accessibility outcomes,
- achievement among students with disabilities,
- use by special-needs learners.

Keywords / cues:
disability, special education, accessibility, inclusive, SEN

Administrative

Changes in school or education-system operations.

Examples:

- processing time,
- prediction accuracy,
- administrative workload,
- resource allocation,
- targeting.

Keywords / cues:
administrative efficiency, processing, allocation, targeting, prediction

---

Other

Use only when none of the above categories fit.

Explain clearly in the outcome name.

---

4. Outcome Name

Definition:
Specific variable or result being measured.

The name should be more precise than the outcome domain.

Examples:

- Mathematics achievement

- Hindi achievement
- Program attendance
- AI-tool take-up
- Teacher questioning quality
- Student dropout
- Weekly AI usage
- Lesson preparation time

Keywords / cues:
dependent variable, primary outcome, secondary outcome, endpoint, measure

---

5. Outcome Measure

Definition:
How the outcome was operationalized or measured.

Examples:

- standardized mathematics test score,
- percentage of available program days attended,
- number of AI sessions per week,
- administrative dropout record,
- teacher observation score,
- self-reported motivation scale.

Be concise.

Keywords / cues:
measured using, constructed from, standardized score, survey scale, administrative records

---

6. Population / Sample for Outcome

Definition:
The group for whom this particular outcome was measured.

This can differ from the overall study sample.

Examples:

- all randomized students,
- treatment students only,

- teachers,
- students who used the AI tool,
- Grade 7 students,
- low-performing subgroup.

Keywords / cues:
analysis sample, respondents, subgroup, treatment users, teachers

---

7. Comparison

Definition:
What the reported result is compared against.

Possible answers:

- Treatment vs control
- Treatment A vs Treatment B
- AI vs human tutoring
- AI vs business-as-usual
- Before vs after
- Users vs non-users
- High usage vs low usage
- Subgroup A vs subgroup B
- No comparison / descriptive outcome

Example:
Students offered Mindspark vs control students not offered free Mindspark access.

Keywords / cues:
compared with, relative to, control, baseline, difference, treatment effect

---

8. Result Value

Definition:
Main numerical estimate reported for the outcome.

Record the number only where practical.

Examples:

- 0.37

- 23
- 58
- 50
- -0.17

Do not include units in this field.

If several estimates are reported, use the principal specification and explain alternatives in `Result Notes`.

Keywords / cues:
coefficient, effect, treatment effect, estimate, mean difference

---

9. Result Unit

Definition:
Unit corresponding to the result value.

Possible answers:

- SD
- percentage points
- percent
- score points
- days
- minutes
- sessions/week
- proportion
- odds ratio
- index points
- currency
- other

Example:
`SD`

Do not confuse percentage points with percent.

---

10. Result Interpretation

Definition:
Short plain-language description of the quantitative result.

Example:
Students offered Mindspark scored 0.37 standard deviations higher in mathematics than control students.

Keep this factual and concise.

Do not add causal interpretation beyond what the study design and authors support.

---

11. Standard Error / Confidence Interval

Definition:
Uncertainty measure reported with the estimate.

Intended answer:
Report either:

- standard error,
- confidence interval,
- or both if available.

Examples:

- SE = 0.07
- 95% CI: 0.12 to 0.32

If not reported:
NR

Keywords / cues:
standard error, SE, confidence interval, CI

---

12. P-value / Statistical Significance

Definition:
Reported p-value or significance information.

Intended answer:
Prefer exact p-value where available.

Examples:

- p = 0.03
- p < 0.01
- statistically significant at 5%
- not statistically significant

If significance is shown only through stars, report the implied threshold if clearly defined by the paper.

Keywords / cues:
p-value, significance, *, **, ***

---

13. Measurement Timing

Definition:
When the outcome was measured relative to the intervention.

Possible answers:

- Endline after 6 weeks
- Endline after 4.5 months
- End of academic year
- Immediately after intervention
- 3-month follow-up
- 1-year follow-up

Keywords / cues:
endline, follow-up, post-treatment, after intervention, months later

---

14. Subgroup

Definition:
Whether the reported outcome applies to a specific subgroup rather than the full sample.

Possible answers:

- Full sample
- Low-performing students
- Girls
- Boys

- Rural students
- Grade 8
- Free-school-meal students
- Low-rated tutors
- Other

If full sample:
Full sample

Keywords / cues:
heterogeneous effects, subgroup, by gender, by baseline achievement, disadvantaged

---

15. Actual vs Intended Usage

Use this field **only for Adoption or Usage outcomes**.

Definition:
Clarifies whether the result refers to:

- planned access,
- initial adoption,
- actual use,
- sustained use.

Possible answers:

- Offered access
- Activated tool
- Used at least once
- Repeat use
- Average usage
- Sustained usage
- Attendance/compliance

Example:
Attendance/compliance: students attended approximately 58% of available program days.

This field is useful because "offered access" is not the same as "actually used."

---

16. Result Notes

Definition:
Short clarification needed to interpret the outcome correctly.

Use for:

- alternative specifications,
- important denominator definitions,
- multiple related estimates,
- whether result is ITT or treatment-on-treated,
- whether result comes from logs or self-reports,
- any ambiguity.

Example:
0.37 SD is the intent-to-treat estimate; IV estimates based on attendance are larger.

Do not use this field for general paper commentary.

Special coding rules

Adoption vs Usage

Do not combine them automatically.

Adoption answers:

> Did the intended user begin using the tool?

Usage answers:

> How much or how often did the user actually use it?

Example:

- 96% tried Khanmigo → `Adoption`
- Median student used Khanmigo on one-third of practice days → `Usage`

Access vs Usage

Do not treat being offered access as actual usage.

Example:

- 500 students received accounts → access
- 350 students logged in → adoption
- average 2.1 sessions/week → usage

Performance vs Learning

If AI helps students perform better while the tool is available, but the study separately measures later unaided learning, record them as separate outcomes.

Example:

- AI-assisted exercise performance → separate outcome
- later unaided test score → Learning

Prediction accuracy

If an AI system predicts dropout risk accurately, record prediction accuracy as an Administrative or Other outcome.

Do not treat prediction accuracy as equivalent to reduced dropout.

If the study also measures actual dropout, create a separate Dropout / Retention outcome row.

Multiple subjects

If a study separately reports mathematics and language outcomes, create separate rows.

Example:

Outcome Domain	Outcome Name
Learning	Mathematics achievement
Learning	Hindi achievement

Multiple time periods

If the same outcome is measured immediately and at longer follow-up, create separate rows when the distinction is important.

Example:

- Mathematics score at endline
- Mathematics score at 12-month follow-up

Output format

Output format

Return one table with one row per distinct outcome.

- Tool ID
- Intervention ID
- Source ID(s)
- Short Citation
- AI Tool / Program
- Outcome Domain
- Outcome Name
- Outcome Measure
- Population / Sample for Outcome
- Comparison
- Result Value
- Result Unit
- Result Interpretation
- Standard Error / Confidence Interval
- P-value / Statistical Significance
- Measurement Timing
- Subgroup
- Actual vs Intended Usage
- Result Notes

Do not merge several substantively different outcomes into one row. Use “Not reported” for unavailable information, “Not verified” for claims that could not be checked, and “NR” for unreported statistical uncertainty or significance as specified above. Numerical examples in this rule are illustrative and must not be entered as study findings.